

AI Credit Risk Intelligence Platform

End-to-End Credit Risk Analytics, Default Prediction and Explainable Decision Support

Presented by

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Business Problem and Platform Goal

- Credit-risk teams need portfolio visibility, applicant-level risk estimates and transparent model explanations in one workflow.
- The objective is to predict probability of default and convert it into business-readable risk scores and Low/Medium/High segments.
- The platform also lets users ask credit-portfolio questions in natural language while enforcing validated, read-only SQL execution.

End User View

Understand portfolio → assess applicant → explain prediction → query business data

Deliverable View

One Streamlit app demonstrating EDA, ML and chatbot in a single end-to-end platform

Evaluation View

Clean pipeline, proper evaluation, explainability, hallucination control and Docker reproducibility

Data Used and Why These Tables Were Chosen

- Application data provides the primary applicant profile, financial variables and the historical default target.
- Bureau data adds external credit-history behaviour such as active/closed credits, debt and overdue exposure.
- Installment data captures repayment behaviour, including payment delay, late-payment rate and payment ratio.

Source Table	Business Role
application_train.csv	Applicant profile, finance and TARGET
application_test.csv	Schema alignment and application records
bureau.csv	Historical credit exposure and overdue behaviour
installments_payments.csv	Repayment delay and payment discipline

Design Principle

Supporting tables are aggregated by SK_ID_CURR before joining, so the analytical data remains one applicant per row.

Applicant-Level Analytical Dataset

Applicants

307,511

Features

140

Duplicates

0

Default Rate

8.07%

Grain

1 row = 1 applicant

- The analytical dataset integrates application, bureau and installment evidence without duplicating applicants.
- Derived features such as credit-to-income, annuity-to-income and employment duration preserve clear credit-risk meaning.
- The dataset is used consistently across EDA, model scoring and the SQLite analytical layer.

Processing Flow

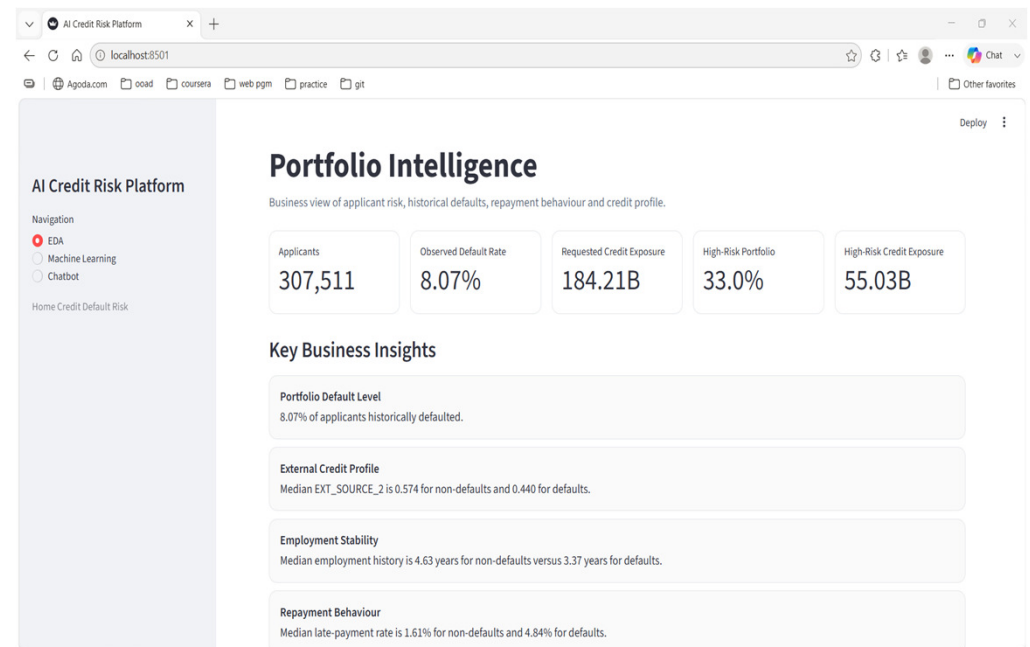
Validate files → clean values → aggregate bureau/installments
→ merge by SK_ID_CURR → engineer ratios → save
analytical dataset

EDA Design: What Was Analysed and Why

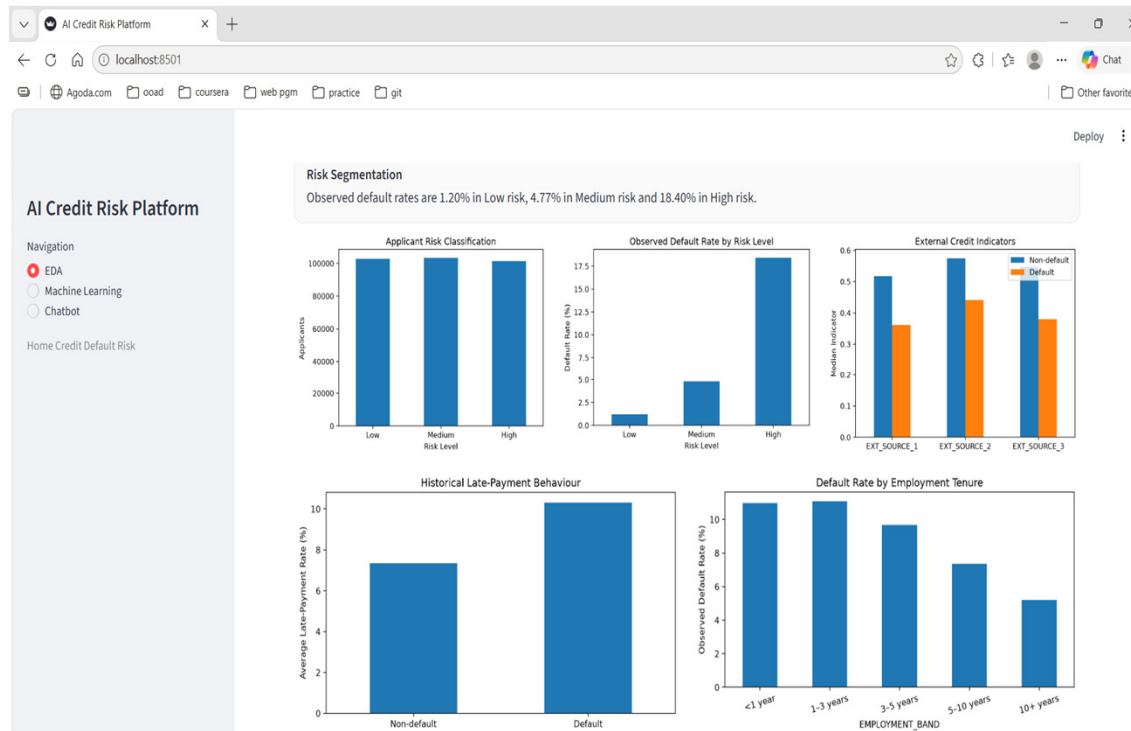
- Default distribution was analyzed first because it determines whether ordinary accuracy is meaningful for this problem.
- External score, employment and repayment behavior were analyzed because they represent major credit-risk dimensions.
- Risk segmentation charts connect model output back to observed portfolio behavior, making the dashboard business-facing.

EDA Interpretation

The dashboard starts with portfolio size, default rate, credit exposure and high-risk exposure so that model outputs are connected to business value.



EDA Charts: Business Meaning



Risk Classification

Shows how the scored portfolio is distributed across Low, Medium and High segments.

Observed Default by Risk

Validates whether higher model risk corresponds to higher historical default behavior.

External Indicators

Shows that defaulters generally have weaker external credit indicators.

Late Payment + Tenure

Connects repayment discipline and employment stability to observed default patterns.

Machine Learning Methodology

- XGBoost was selected because the data is structured, non-linear, mixed-type and contains meaningful feature interactions.
- Class imbalance is handled through stratified splits and dynamic scale_pos_weight rather than synthetic applicant generation.
- The 70/10/20 split separates model training, calibration/threshold selection and final untouched test evaluation.

Model

XGBoost Classifier

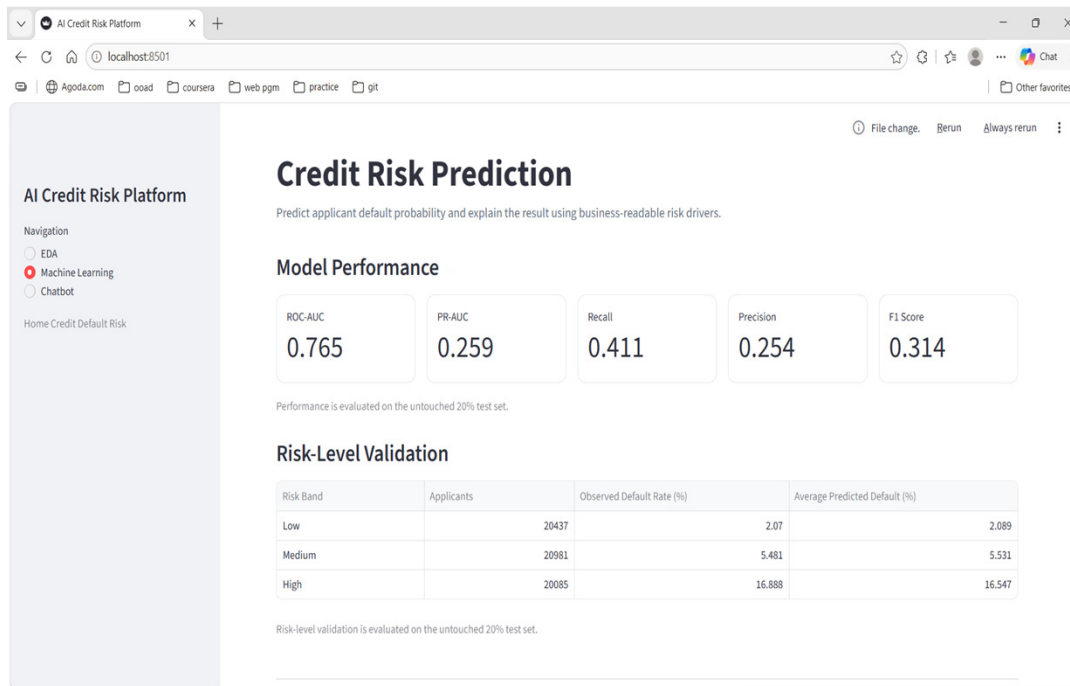
Calibration

Platt / Sigmoid calibration for user-facing probabilities

Operating Point

Threshold 0.1562 selected only on calibration data by maximizing F1

Model Performance and Risk Validation



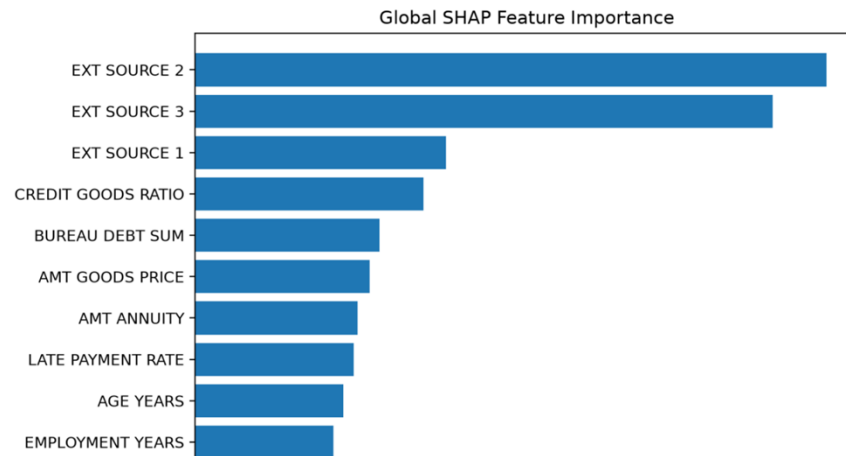
- ROC-AUC 0.7645 shows useful ranking quality between default and non-default applicants.
- PR-AUC 0.2586 and F1 0.3140 demonstrate meaningful default-class detection under class imbalance.
- Risk bands are validated on the untouched test set, with observed default rising from 2.07% to 16.89%.

Risk	Observed	Predicted
Low	2.070%	2.089%
Medium	5.481%	5.531%
High	16.888%	16.547%

Applicant-Level Risk Assessment

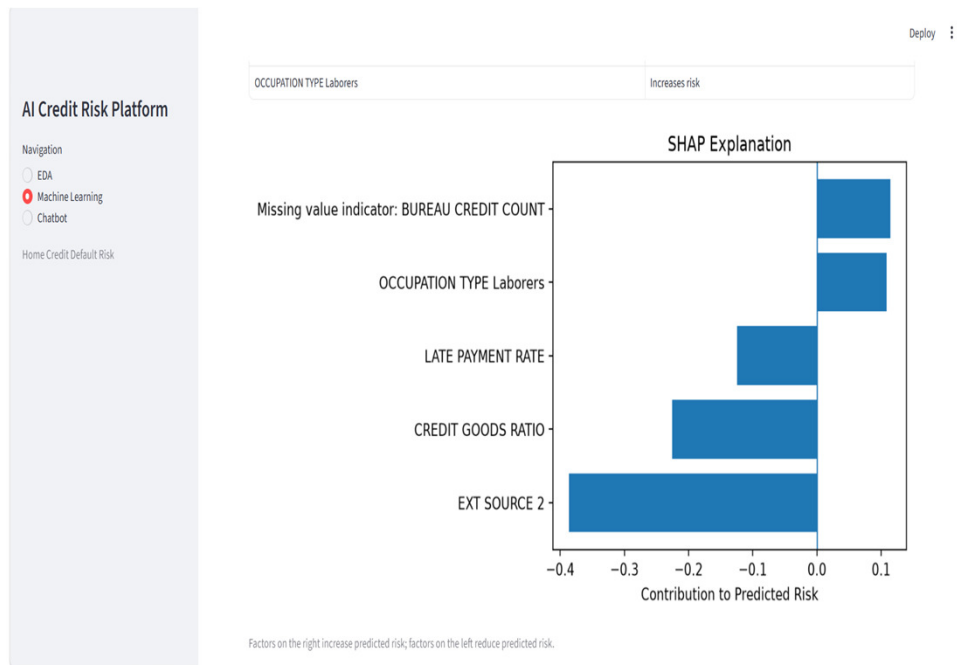
Deploy

Key Risk Drivers Across the Portfolio



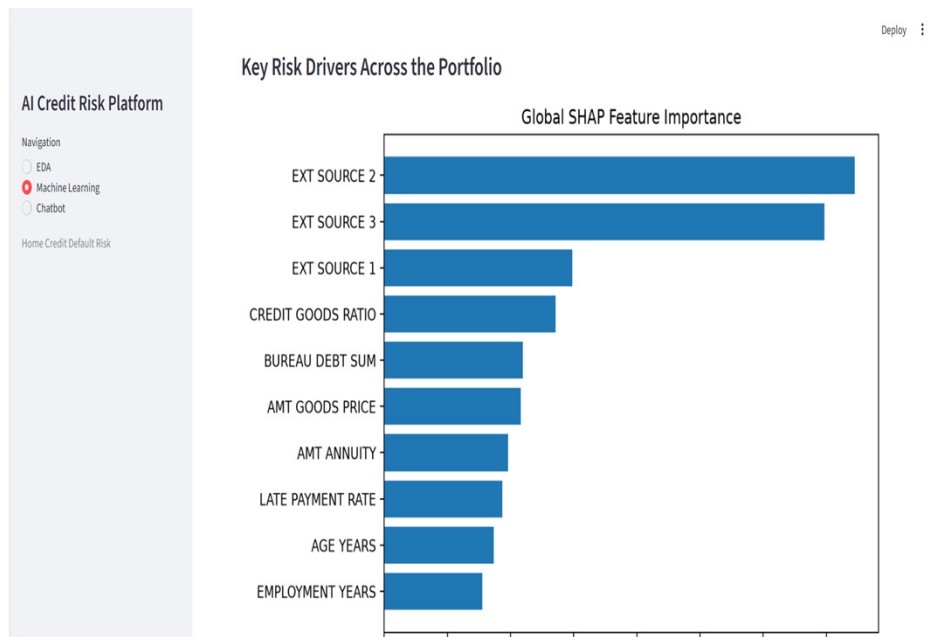
- A user enters `SK_ID_CURR` to retrieve an applicant already present in the analytical dataset.
- The model returns calibrated probability of default, a 0–100 risk score and a Low/Medium/High classification.
- Applicant profile fields provide context for the prediction without exposing unnecessary technical details.

Explainable AI: Why This Prediction?



- Local SHAP decomposes an individual prediction into factors that increase or reduce predicted risk.
- For the shown applicant, EXT_SOURCE_2 and credit-to-goods ratio reduce predicted risk, while missing bureau history increases risk.
- The explanation is presented as business-readable risk drivers rather than raw model internals.

Portfolio-Level Explainability and Model Rules



- Global SHAP identifies which features have the strongest overall influence on predictions across the portfolio.
- External credit indicators dominate the global explanation, followed by financial burden and repayment-history features.
- ML-derived rules summarize recurring model patterns for business review while XGBoost remains the prediction model.

ML-Derived Business Rules

Rule 1

When $EXT_SOURCE_3 > 0.309$ AND $EXT_SOURCE_2 > 0.406$ AND $EXT_SOURCE_3 \leq 0.536$

Approximate default probability: 6.44%

Rule 2

When $EXT_SOURCE_3 > 0.309$ AND $EXT_SOURCE_2 > 0.406$ AND $EXT_SOURCE_3 > 0.536$

Approximate default probability: 3.31%

Rule 3

When $EXT_SOURCE_3 > 0.309$ AND $EXT_SOURCE_2 \leq 0.406$ AND $EXT_SOURCE_3 \leq 0.536$

Approximate default probability: 14.57%

Talk-to-Data: Natural-Language Portfolio Analytics

The screenshot shows a web browser window with the URL `localhost:8501`. The page title is "AI Credit Risk Platform". On the left, there is a navigation sidebar with the following items: "Navigation", "EDA", "Machine Learning", "Chatbot" (selected), and "Home Credit Default Risk". The main content area displays a query input field with the text "Compare historical late-payment behaviour across risk bands." and a red "Ask" button. Below the input, the "Answer" section contains a text summary: "Historical late-payment behavior aligns with the model-predicted risk bands, showing higher observed late-payment rates and average payment delays as predicted risk increases. Observed late-payment rates rise progressively from 4.38% in the Low risk band to 7.62% in Medium and 10.65% in High, with average payment delays increasing correspondingly from 0.74 to 1.06 and 1.39 days. Average maximum payment delay is lowest for Low risk applicants (15.15 days), while Medium risk applicants recorded the highest average maximum delay at 20.18 days, slightly above High risk applicants at 19.69 days." Below the text is a table with the following data:

RISK_BAND	applicants	avg_late_payment_rate_pct	avg_payment_delay	avg_max_payment_delay
Low	102839	4.38	0.74	15.15
Medium	103290	7.62	1.06	20.18
High	101382	10.65	1.39	19.69

Below the table, there is a section titled "View Generated SQL" which displays the following SQL query:

```
SELECT RISK_BAND, COUNT(*) AS applicants, ROUND(AVG(LATE_PAYMENT_RATE) * 100, 2) AS avg_late_payment_rate_pct, ROUND(AVG
```

- Users can ask business questions without writing SQL, such as comparing repayment behavior across risk bands.
- The system converts the question into SQL, validates it, executes it against SQLite and summarizes the returned result.
- Displayed SQL improves auditability, making the chatbot transparent instead of a black-box answer generator.

SQL Safety and Core-Query Resilience

The screenshot shows a web browser window titled "AI Credit Risk Platform" at the URL "localhost:8501". The interface includes a sidebar with navigation options: EDA, Machine Learning, and Chatbot (selected). The main content area displays a chatbot response: "Average historical late-payment rates by risk band are Low: 4.38%, Medium: 7.62%, High: 10.65%." Below this, a table shows the data:

RISK_BAND	average_late_payment_rate_pct
Low	4.38
Medium	7.62
High	10.65

Below the table, a section titled "View Generated SQL" shows the following SQL query:

```
SELECT
  RISK_BAND,
  ROUND(AVG(LATE_PAYMENT_RATE) * 100, 2)
  AS average_late_payment_rate_pct
FROM credit_applicants
GROUP BY RISK_BAND
ORDER BY
  CASE RISK_BAND
    WHEN 'Low' THEN 1
    WHEN 'Medium' THEN 2
    WHEN 'High' THEN 3
    ELSE 4
  END
```

- Generated SQL is restricted to validated SELECT queries over approved analytical columns.
- The five predefined business questions also have deterministic SQL templates for consistent demo-critical answers.
- This design combines flexible NL-to-SQL for custom questions with reliable execution for core portfolio analysis.

Controls

Schema grounding • SQL validation • read-only execution • bounded repair • deterministic templates

Application Packaging and Deployment

- The full platform is demonstrated through a lightweight Streamlit interface with EDA, Machine Learning and Chatbot sections.
- Docker Compose packages the runtime so an evaluator can run the application with a single command.
- Model artifacts are saved in models/, while raw datasets and environment secrets are excluded from GitHub.

Run Locally

```
streamlit run app.py
```

Run with Docker

```
docker compose up --build
```

Required Env

```
GEMINI_API_KEY, LLM_MODEL
```

Repository

```
AI-Credit-Risk-Platform
```

Final Outcome

- Built a complete credit-risk intelligence platform mapped to EDA, ML, XAI, Talk-to-Data, UI and Docker requirements.
- Delivered calibrated default probabilities, explainable applicant-level predictions and test-validated Low/Medium/High risk bands.
- Created a professional, reproducible application with GitHub-ready structure, Docker deployment and presentation-ready outputs.

ML Result

ROC-AUC 0.7645 • PR-AUC 0.2586 • F1 0.3140

Risk Validation

Low 2.07% • Medium 5.48% • High 16.89%

Platform Result

EDA + ML + SHAP + Chatbot + Docker in one app